Example-2

Polynomial Kernel (of order 2)

$\lfloor u \quad v \top 1 \rfloor$

a_1, v_1

a_2, v_2

Map Input data from 2-D to 6-D

Normal Model — Calculate $u \quad v$

$\lfloor a_1 \quad a_2 \rfloor$
$b_1 \quad b_2$

a_1, v_1

a_2, v_2

Kernelized Model — Calculate $u \quad v$

$\lfloor a_1, a_2, \vee z a_1 a_2, \vee z a_1, \vee z a_2, 1 \rfloor$

$\begin{pmatrix} \\ \\ \end{pmatrix}$
b_1^2
b_2^2
$\sqrt{2} b_1 b_2$
$\sqrt{2} b_1$
$\sqrt{2} b_2$
1

Higher Computational Complexity!!!

Vectors Mapping

$\begin{pmatrix} a_1^2 \\ \cdots \\ 1 \end{pmatrix}$

$\begin{pmatrix} b_1^2 \\ \cdots \\ 1 \end{pmatrix}$